

Planar Kepler Conics, Collision Incompleteness, and the Levi-Civita Configuration Boundary

Route-A structural certificate HCS-C216

Abstract

We close a convention-locked theorem package for the attractive planar Kepler Hamiltonian. Runge-Lenz identities classify every nonzero-angular-momentum orbit as an ellipse, parabola, or hyperbola; the negative-energy period and radial action and the positive-energy scattering angle are normalized exactly. The zero-angular-momentum branch reaches the removed collision in finite physical time. Levi-Civita’s fixed-energy change of variables gives a smooth configuration equation, while a time-period resonance fixes a continuum rather than isolated primitive orbits.

Keywords: Kepler problem; Runge-Lenz vector; collision regularization; radial action; fixed-point continuum; Route A.

1 Frozen Hamiltonian and invariants

Identify the plane with $\mathbb{C}$ only when the Levi-Civita map is introduced. For $q \in \mathbb{R}^2 \setminus \{0\}$, $p \in \mathbb{R}^2$, and $\mu > 0$, set

$$H(q, p) = \frac{|p|^2}{2} - \frac{\mu}{r}, \quad r = |q|.$$

Hamilton’s equations are $\dot{q} = p$ and $\dot{p} = -\mu q/r^3$. Along a collision-free solution define

$$E = H, \quad L = q_1 p_2 - q_2 p_1, \quad A = (|p|^2 - \mu/r)q - (q \cdot p)p. \quad (1)$$

Differentiation gives $\dot{E} = \dot{L} = \dot{A} = 0$. Taking the scalar product and norm of A gives the two identities

$$A \cdot q = L^2 - \mu r, \quad |A|^2 = \mu^2 + 2EL^2. \quad (2)$$

2 All-energy conics, period, action, and scattering

Suppose $L \neq 0$, let $e = |A|/\mu$, and measure θ from A . Equation (2) is equivalent to

$$r(\theta) = \frac{L^2/\mu}{1 + e \cos \theta}, \quad e^2 - 1 = \frac{2EL^2}{\mu^2}. \quad (3)$$

Thus $E < 0$ is an ellipse (a circle when $e = 0$), $E = 0$ a parabola, and $E > 0$ a hyperbola. In the bound case put $a = -\mu/(2E)$ and $r_{\pm} = a(1 \pm e)$. Kepler’s period and the radial momentum are

$$P(E) = 2\pi\mu(-2E)^{-3/2}, \quad p_r^2 = 2\left(E + \frac{\mu}{r}\right) - \frac{L^2}{r^2}. \quad (4)$$

The normalization is part of the theorem, not a convention left implicit:

$$J_r = \frac{1}{2\pi} \oint p_r \, dr = \frac{1}{\pi} \int_{r_-}^{r_+} p_r \, dr = \frac{\mu}{\sqrt{-2E}} - |L|, \quad E < 0. \quad (5)$$

The turning-point substitution also gives $\partial_E J_r = P(E)/(2\pi)$. For $E > 0$, the asymptotic directions of (3) have $\cos \theta_{\infty} = -1/e$, so the unsigned hyperbolic deflection is

$$\chi = 2 \arcsin(1/e), \quad e > 1. \quad (6)$$

Revision focus. Round 0 freezes the Hamiltonian, invariants, conic equation, and energy-sign classification.

Declarations

Data and code. Synthetic exact probes and package-local deterministic code accompany the manuscript; no training data are used. **Ethics/funding.** No human or animal data; no external funding reported. **AI disclosure.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.